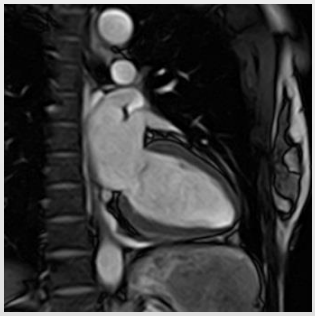
＊“mCardiac package” is required.

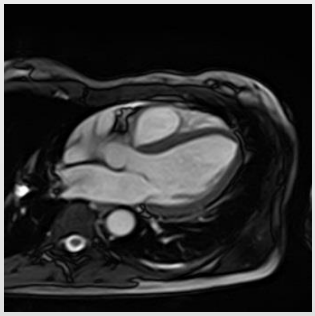
Cardiac MR : Planning

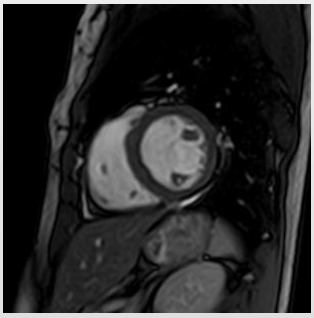
Introduction

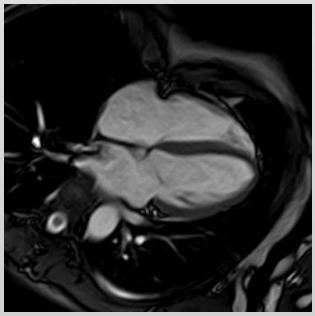
Cardiac MRI has the following basic cross sections.
・LV two chamber(Vertical long)
・Horizontal long
・LV short Axial
・Four chamber
・Three chamber
This section describes how to set each cross section.

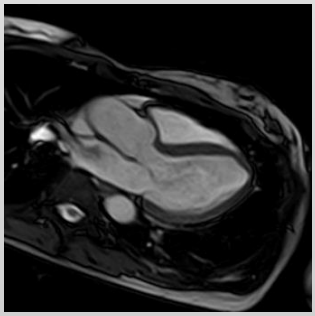
Repeat imaging multiple times to get any basic cross section.
Explain to the patient that they should hold their breath at the same position each time.
Also, the position, shape, and axis of the heart vary depending on the disease and body habitus.
Please set the cross section while observing multiple images to obtain the basic cross section correctly.


二腔長軸像（垂直長軸像）


水平長軸像


左室短軸像

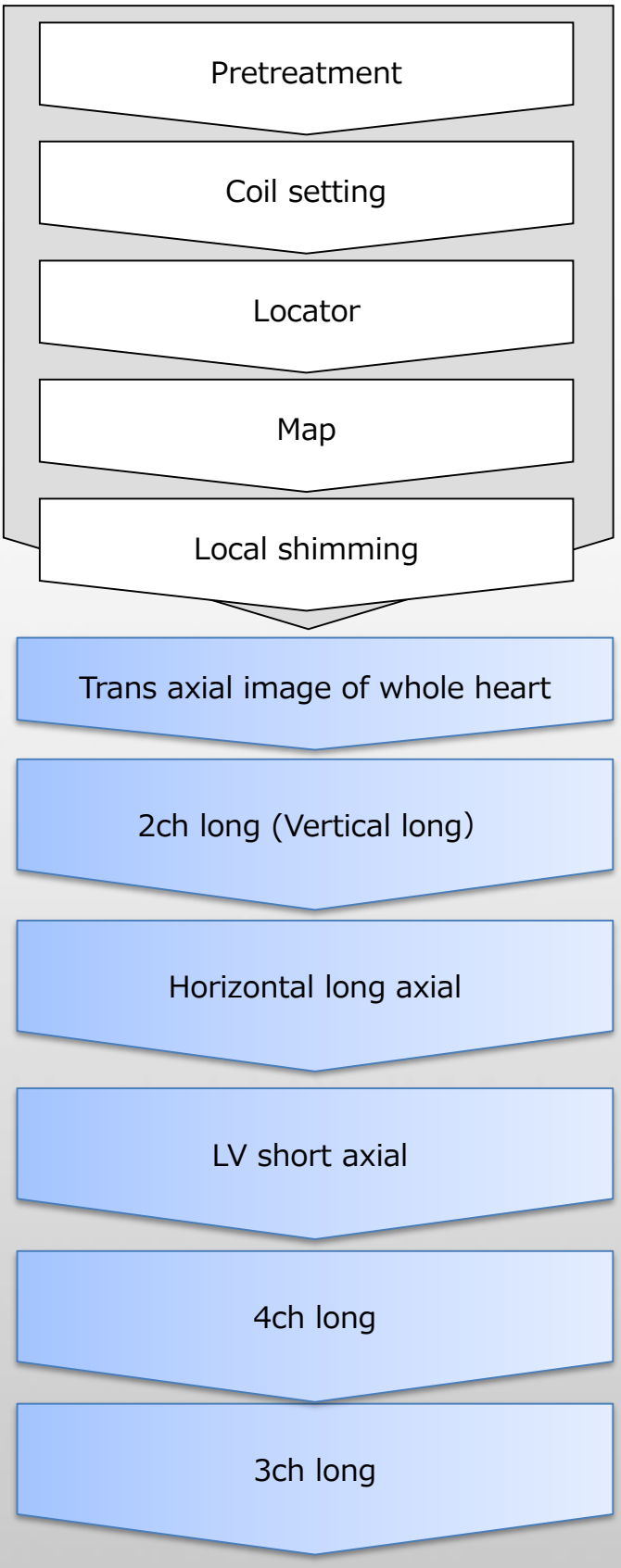

四腔長軸像


三腔長軸像

imaging conditions

・All images are taken with ECG and also breath-hold.

scanning workflow



Planning trans axial images that covers the whole heart.

LV two chamber (vertical) long axis prescribed orthogonal to trans axial scouts aligned through the apex and center of the mitral valve.

Four chamber (horizontal) long axis aligned orthogonal to the 2 chamber long axis, passing through the center of the mitral valve and left atrium and continuing through the long axis of the LV.

The first short-axis cine plane should be planned using the 4- and 2-chamber long-axis views.

The 4-chamber long-axis view is prescribed from the 2-chamber long-axis view through the apex and center of the mitral and tricuspid valves.

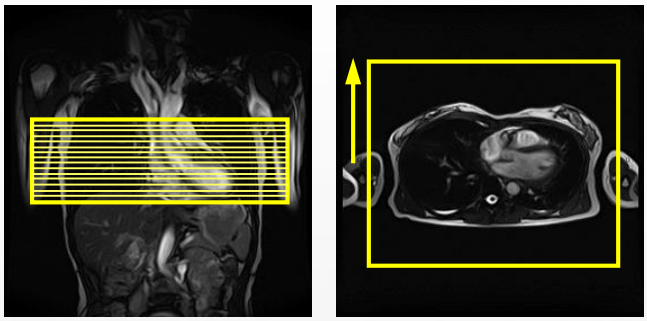
The 3-chamber LV view is prescribed passing through the apex, the center of the mitral valve and aligned with the center of LV outflow tract (LVOT) to aortic valve, as seen on a basal short axis cine.

●scanning workflow

Trans Axial

Planning trans axial images that covers the whole heart.

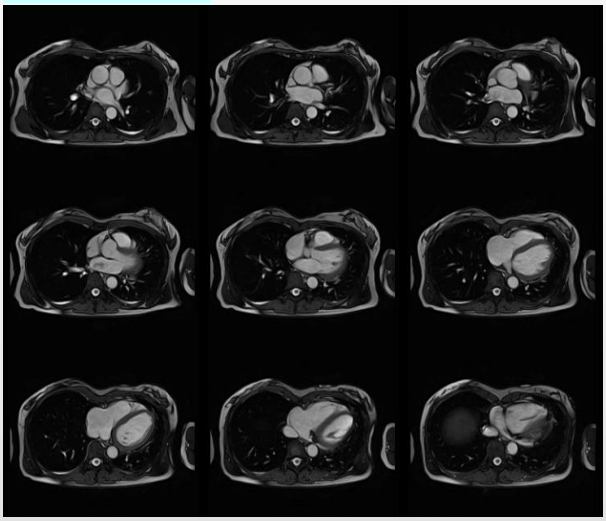
（1） Adjust the number of slice and FOV to cover the entire heart.



（2） Set the gate condition.
Read the patient's R-R interval / heart rate.

（3）） Take an image with a breath-hold.
Explain breath-hold imaging to the patient.
Since breath-hold imaging is repeated multiple times, ask the patient to hold your breath at the same position each time.

Trans axial

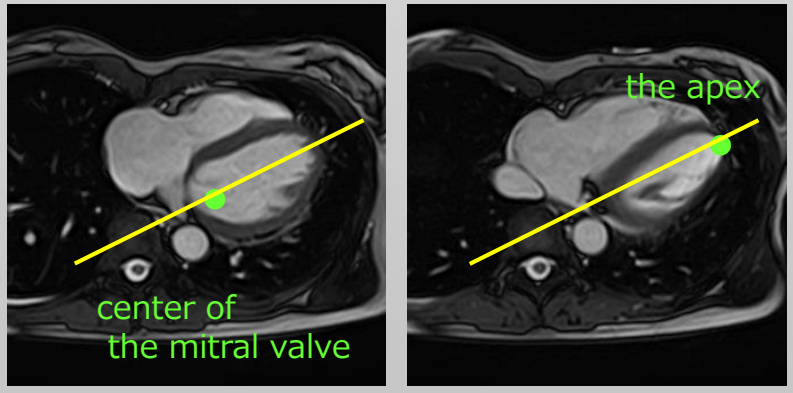


trueSSFP法

2ch long (Vertical long)

LV two chamber (vertical) long axis prescribed orthogonal to trans axial scouts aligned through the apex and center of the mitral valve.

（1） Based on the imaging conditions of the AX image of the entire heart, adjust the number of slices (3 to 5) and FOV, and set the encoding direction so that there is no aliasing.



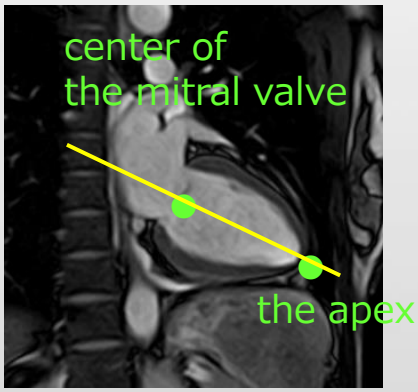
(2) Take an image with a breath-hold.

2ch long

Horizontal long

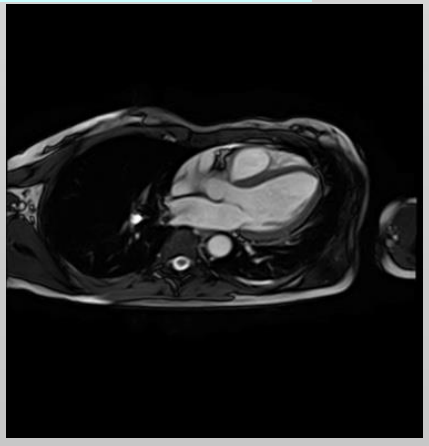
LV two chamber (vertical) long axis prescribed orthogonal to trans axial scouts aligned through the apex and center of the mitral valve.

(1) Based on the imaging conditions of the AX image of the entire heart, adjust the number of slices (3 to 5) and FOV, and set the encoding direction so that there is no aliasing.



(2) Take an image with a breath-hold.

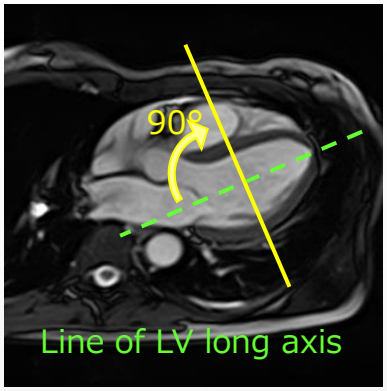
Horizontal long



LV short Ax

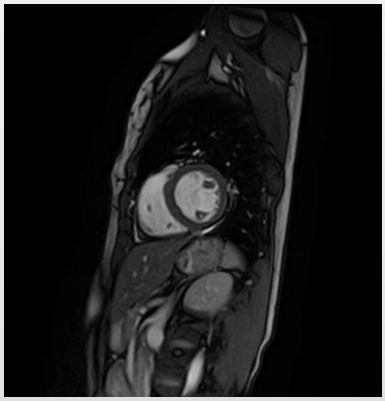
The first short-axis cine plane should be planned using the 4- and 2-chamber long-axis views.

(1) Based on the imaging conditions of the AX image of the entire heart, adjust the number of slices (3 to 5) and FOV, and set the encoding direction so that there is no aliasing.



(2) Take an image with a breath-hold.

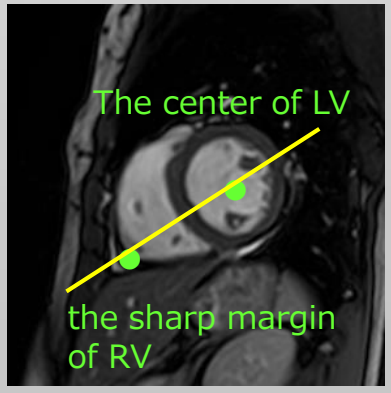
LV short



4ch long

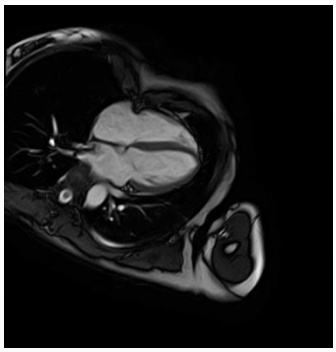
The 4-chamber long-axis view is prescribed through the sharp margin of the free wall of the right ventricle (RV) and the center of the LV.

(1) Based on the imaging conditions of the AX image of the entire heart, adjust the number of slices (3 to 5) and FOV, and set the encoding direction so that there is no aliasing.



(2) Take an image with a breath-hold.

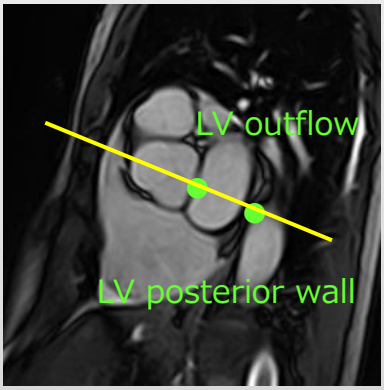
4ch long



3ch long

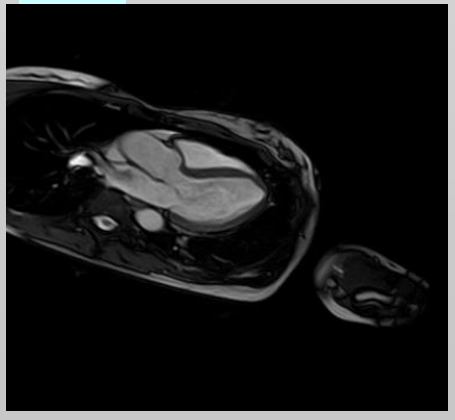
The 3-chamber LV view is prescribed passing through the left ventricular outflow tract and the posterior wall on the left ventricular short-axis image.

(1) Based on the imaging conditions of the AX image of the entire heart, adjust the number of slices (3 to 5) and FOV, and set the encoding direction so that there is no aliasing.



(2) Take an image with a breath-hold.

3 ch



scan parameter

true SSFP	1.5T	3T
Scan Time	Around 0:03 (Depends on HR and Num. of slice)	Around 0:03 (Depends on HR and Num. of slice)
Sequence	SSFP2D	SSFP2D
TR	4.2	3.4
TE	2.1	1.7
Plane	AX	AX
Encode Direction	AP	AP
Num. of slice	Required	Required
Slice thickness	8.0/2.0	7.0/1.4
Matrix	192×256	192×256
FOV	36×36	36×36
No PE Wrap	1.0/1.0	1.0/1.0
SPEEDER	2.0	2.0
NAQ	1	1
Flip Angle	70	70
FatSAT	Off	Off
Num. of Segment/Type	2/Sequential	2/Sequential
Gating Source	ECG	ECG
Delay Time	1	1
Delay time	Set the Acquire window to be in diastole	Set the Acquire window to be in diastole

Provided by "Canon Medical Systems MRI application support service"